

homework-2

Packages

Question 1

```
install.packages("aRtsy")
```

`install.packages()` is used to install a package on the computer, while `library()` is used to load an installed package into the current R session. Usually, a package only needs to be installed once, but `library()` needs to be run every time when starting a new R session and want to use that package.

Question 2

```
library(aRtsy)
```

The Collatz conjecture is a mathematical problem involving positive integers. If a number is even, divide it by 2; if it is odd, multiply it by 3 and add 1. The conjecture states that repeating this process will eventually reach 1 for any positive integer.

The Collatz equation is $f(n) = n/2$, while n is even, $f(n) = 3n + 1$, while n is odd.